

Name:

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MANIPAL UNIVERSITY
JAIPUR

Odd Semester Mid Term Examination, Academic Session: Jul-Nov 2026

FoSTA, School of Computer Science & Engineering
Department of Artificial Intelligence & Machine Learning
B.Tech. - Artificial Intelligence & Machine Learning

Course Code: AIM3144

Course: High Performance Computing

Semester: Vth

Time: 1.5 hrs.

Max. Marks: 30

Instructions: All questions are compulsory.

Missing data, if any, may be assumed suitably.

Calculator is not allowed.

SECTION A

S.No.	SECTION A		Marks	CO										
Q 1	Match the following: <table><tr><th>Instructions</th><th>Addressing Modes</th></tr><tr><td>1. ADD R1, R2, R3</td><td>A. Immediate Addressing</td></tr><tr><td>2. LW R4, 8(R2)</td><td>B. Register Addressing</td></tr><tr><td>3. ADDI R5, R4, 5</td><td>C. Base-Displacement Addressing</td></tr><tr><td>4. SW R6, 16(R3)</td><td>D. Base-Displacement Addressing</td></tr></table>		Instructions	Addressing Modes	1. ADD R1, R2, R3	A. Immediate Addressing	2. LW R4, 8(R2)	B. Register Addressing	3. ADDI R5, R4, 5	C. Base-Displacement Addressing	4. SW R6, 16(R3)	D. Base-Displacement Addressing	(2)	CO1
Instructions	Addressing Modes													
1. ADD R1, R2, R3	A. Immediate Addressing													
2. LW R4, 8(R2)	B. Register Addressing													
3. ADDI R5, R4, 5	C. Base-Displacement Addressing													
4. SW R6, 16(R3)	D. Base-Displacement Addressing													
Q 2	State which of the following is a branch instruction in MIPS? a) LW b) SW c) BEQ d) JAL		(1)	CO2										
Q 3	Identify the MIPS instruction which is typically used to return control to the caller function? a) J b) JR \$ra c) JAL d) BNE		(1)	CO3										
Q 4	Recall different types of parallel computers with one line description for all of them.		(2)	CO1										

SECTION B

Q 5	Evaluate the MIPS1 branch instruction set architecture. What are the different special purpose registers used in MIPS.	(5)	CO2						
Q 6	Contrast between LB (load byte) and LBU (load byte unsigned) in MIPS1. Explain with examples and instructions.	(5)	CO1						
Q 7	Consider the following MIPS instructions that load and store data in memory. Assume the system uses Big-endian byte ordering. LBU R1, 0(R2) LB R3, 1(R2) Assume register R2 = 0x5000 and the memory content starting from 0x5000 is: <table border="1"><tr><th>Address</th><th>Value (Hex)</th></tr><tr><td>0x5000</td><td>0x7F</td></tr><tr><td>0x5001</td><td>0xA0</td></tr></table> a) Identify the value which will be loaded into R1 after the first instruction (LBU R1, 0(R2)) executes?	Address	Value (Hex)	0x5000	0x7F	0x5001	0xA0	(5)	CO3
Address	Value (Hex)								
0x5000	0x7F								
0x5001	0xA0								

SECTION-C

Q 8	Given the following MIPS instructions: ADD R1, R2, R3 Instruction 1; LW R4, 8(R1) Instruction 2 SW R5, 12(R1) Instruction 3; JR R6 Instruction 4		
a)	Calculate the total number of cycles required to execute the ADD instruction? Explain each stage.	(3)	CO1
b)	Calculate the total number of cycles required to execute the SW (Store Word) instruction? Explain each stage.	(3)	CO1
c)	Calculate the total number of cycles required to execute the JR (Jump Register) instruction? Explain each stage.	(3)	CO2